

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech Mid Semester Exam

Semester: 6th**Subject Code: 303105311****Subject Name: Quant & Reasoning****Date: 27/01/2025****Time: (1hr: 30min)****Total Marks: 40**

Sr. No.		Marks	Co/Po	Blooms Taxonomy
Q.1	(A) Five One line Questions 1) Find H.C.F(126, 162, 180) 2) Is 2457 divisible by 7? 3) If the sum is 240 and average is 40, find the number of quantities. 4) If 40% of an amount is 250, what will be 60% of that amount? 5) What is the formula of mean proportional?	05 01 01 01 01 01	 2 2 1 1 2	 Apply Apply Evaluate Evaluate Apply
	(B) Five Fill in the blanks 1) The Comparison of two quantities is known as ____? 2) Probability of an impossible event is ____? 3) ${}^6C_4 = ___?$ 4) How many cardinal directions are there? 5) The n^{th} term of harmonic progression is given by ____?	05 01 01 01 01 01	 2 2 2 2 2	 Apply Apply Apply Apply Apply
Q.2	Attempt any four(Short Questions)	12		
	(1) In Jack's room, Jack has three trousers. One of them is black, the second is blue, and the third is brown. In his room, he also has four shirts. One of them is black and the other 3 are white. He opens his room in the dark and picks out one shirt and one trouser pair without examining the colour. What is the probability that neither the shirt nor the trouser is black?	03	3	Analyze
	(2) Y is in the East of X which is in the North of Z. If P is in the South of Z, then in which direction of Y, is P?	03	1	Evaluate
	(3) The ratio of the number of boys and girls in a college is 7:8. If the percentage increase in the number of boys and girls be 20% and 10% respectively, what will be the new ratio?	03	2	Apply
	(4) How many 3 digit number can be formed with the digits 5, 6, 2, 3, 7 and 9 which are divisible by 5 and none of its digit is repeated?	03	1	Evaluate
	(5) 8th term of the sequence $2\sqrt{4}, \sqrt{4}, 0, \dots$ Will be?	03	2	Apply
Q.3	Attempt any two questions	08		

	(1) Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:	04	1	Evaluate
	(2) Five years ago, A was thrice as old as B and ten years later, A shall be twice as old as B. What are the present ages of A and B?	04	3	Analyze
	(3) How many integers, greater 999 but not greater than 4000, can be formed with the digit 0, 1, 2, 3 and 4, if repetition of digits is allowed?	04	1	Evaluate
Q.4	(A) The number of students in 3 classes is in the ratio of 2 : 3 : 4. If 12 students are increased in each class this ratio changes to 8 : 11 : 14. The total number of students in the three classes in the beginning was	05	2	Apply
	(B) Eleven friends M, N, O, P, Q, R, S, T, U, V and W are sitting in the first row of the stadium watching a cricket match. T is to the immediate left of P and third to the right of U. V is the immediate neighbour of M and N and third to the left of S. M is the second to the right of Q, who is at one of the ends. R is sitting next to the right of P and P is second to the right of O. Which people are sitting to the right of S?	05	1	Evaluate
	OR			
	(B) Eight friends M, N, O, P, Q, R, S and T are sitting around a circular table and belong to eight different states Goa, Tamilnadu, Telangana, Karnataka, Rajasthan, Jharkhand, West-Bengal and Madhya Pradesh. They read three Newspapers in The Hindu, Times of India and Indian Express. At least two persons read one newspaper not necessarily in the same order. Only P and Q read The Hindu Newspaper. P is third to the right of Q. M is from West Bengal reads The Times of India Newspaper and is sitting immediately right of Q. R is from Karnataka and reads Indian Express Newspaper. T reads the same Newspaper as S reads who is third to the left of R and fourth to the right of N. O is sitting exactly opposite of P who belongs to Goa. Persons belonging to Madhya Pradesh and Telangana read the Time of India Newspaper. S is from Jharkhand while N is from Telangana and reads the Times of India Newspaper. T is from Rajasthan. From which state Q belongs?	05	1	Evaluate